



RESEARCH · THEORY OF CONSTRAINTS · CAUSAL GNN

The Cloud Paradox

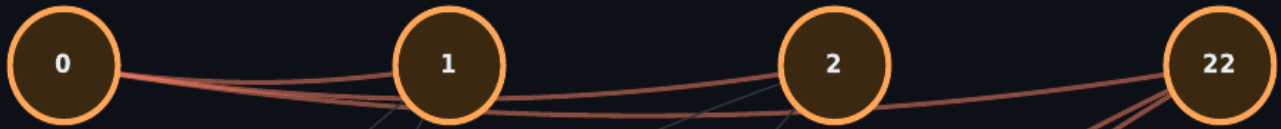
**Why is your system
FASTEST right before it crashes?**

A causal AI model that learned the answer
from 3.3 million real distributed traces.

Swipe for the architecture →

#MachineLearning #GNN #SRE

1 · ENTRY & FRONTEND



2 · ORCHESTRATION & LOGIC



3 · STORAGE & CACHING CORE



4 · PERIPHERAL & HELPERS



post-storage-mongodb
The irreducible core

Pattern D — Entry layer Pattern D — Storage core Other services (24 nodes)

Mapping 3.3 Million Traces into 7 Deterministic Structural Patterns

Pattern D: Entry + Storage Core
49.7% of all bottleneck experiments (88 of 177 files)

30 nodes · 17 servers · DeathStarBench

Real numbers. No inflation.

0.869

Detection AUC

5-fold val · compose graph

88.9%

Pattern Accuracy

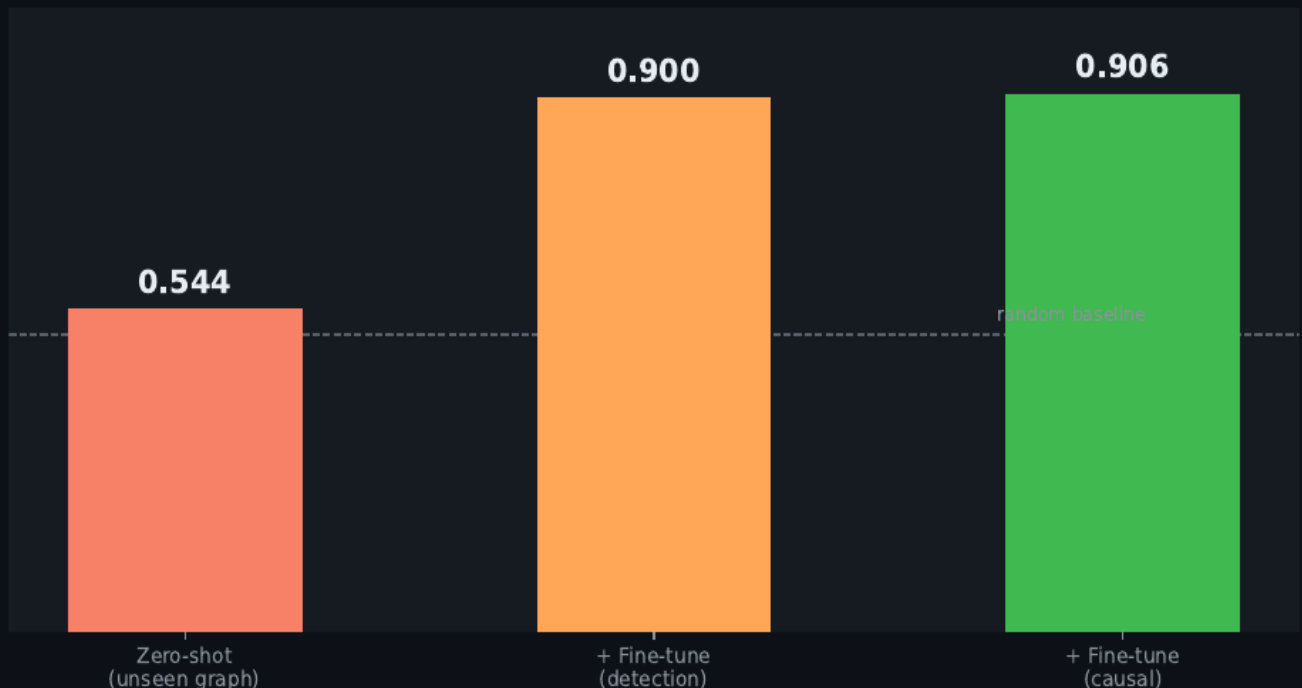
8-class taxonomy (A-G + none)

0.97

Root-Cause Recall

vs 0.94 random baseline

**Generalization to an UNSEEN 7-node graph (Home workflow)
AUC before vs after adaptation**



Zero-shot transfer failed (0.544) —

~25 minutes of fine-tuning recovered AUC = 0.906

Full causal root-cause analysis & paper — coming soon